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IDE Used	Visual Studio 2026
GitHub Repository URL	https://github.com/TiannaD/ShippingCharges.git

Problem Restatement

This program takes the package weight and shipping distance as user input and calculates the total shipping cost. The total shipping cost is calculated using the package weight and shipping distance within a valid range to get the rate multiplier. Packages less than zero and greater than 20kg are invalid, and shipping distances less than 10 or greater than 3000 are also invalid.

Required Calculations

- Weight ranges and rates:
    - If package 2kg or less -> rate = $1.10
    - If package greater than 2kg but less than or equal to 6kg -> rate = $2.20
    - If package greater than 6kg but less than or equal to 10kg -> rate = $3.70
    - If package greater than 10kg but less than or equal to 20kg -> rate = $4.80
- Expression to determine the number of 500 mile units
    - Multiplier = ceil(shipping distance/ 500)
- Expression to calculate total shipping cost
    - Total cost = rate * multiplier
- Ordinary truncation would produce incorrect results for distances that aren't multiplies of 500 because they would provide a multiplier as a double and wouldn't round up when needed.

Algorithm

1. Start the program
2. Declare const int of PACKAGE_MIN = 0, PACKAGE_MAX = 20, SHIPPING_MIN = 10, SHIPPING_MAX = 3000, PER_MILE = 500
3. Declare double of rate, multiplier, totalCost, packageWeight, and shippingDistance
4. Output user prompt, "Enter package weight (kg)", store response in packageWeight variable

5. Output user prompt, "Enter shipping distance", store response in shippingDistance variable
6. Check if packageWeight > PACKAGE _MIN and packageWeight <= PACKAGE_MAX. Else if packageWeight < PACKAGE_MIN, output "Invalid package weight." Else if packageWeight > PACKAGE_MAX, output "Package exceeds weight limit."
7. Check if shippingDistance is >= SHIPPING_MIN and shippingDistance <= SHIPPING_MAX. Else if shippingDistance is < SHIPPING_MIN and shippingDistance > SHIPPING_MAX, output "Invalid shipping distance."
8. Check if packageWeight <= 2, rate = 1.1, multiplier = ceil(shippingDistance / PER_MILE), totalCost = rate * multiplier, output
9. End program

Pseudocode

START

DECLARE const int PACKAGE MIN= 0

DECLARE const int PACKAGE MAX = 20

DECLARE const int PACKAGE MIN= 10

DECLARE const int SHIPPING MAX = 3000

DECLARE const int PER MILE = 500

DECLARE rate

DECLARE multiplier

DECLARE totalCost

DECLARE packageWeight

DECLARE shippingDistance

OUTPUT "Enter package weight(kg): "

DEFINE user input = packageWeight

OUTPUT "Enter shipping distance(mi): "

DEFINE user input = shippingDistance

IF (packageWeight > PACKAGE_MIN and packageWeight <= PACKAGE_MAX)

    IF (shippingDance >= SHIPPING_MIN and shippingDistance <= SHIPPING MAX)

IF (packageWeight <= 2)

rate = 1.1

multiplier = Ceil(shippingDistance / PER_MILE)

totalCost = rate * multiplier

OUTPUT "Weight: ", packageWeight, "Distance: ", shippingDistance, "Total Shipping: $", totalCost

ELSE IF (packageWeight > 2 and packageWeight <= 6)

rate = 2.2

multiplier = Ceil(shippingDistance / PER_MILE)

totalCost = rate * multiplier

OUTPUT "Weight: ", packageWeight, "Distance: ", shippingDistance, "Total Shipping: $", totalCost

ELSE IF (packageWeight > 6 and packageWeight <= 10)

rate = 3.7

multiplier = Ceil(shippingDistance / PER_MILE)

totalCost = rate * multiplier

OUTPUT "Weight: ", packageWeight, "Distance: ", shippingDistance, "Total Shipping: $", totalCost

ELSE IF (packageWeight > 10 and packageWeight <=20)

rate = 4.8

multiplier = Ceil(shippingDistance / PER_MILE)

totalCost = rate * multiplier

OUTPUT "Weight: ", packageWeight, "Distance: ", shippingDistance, "Total Shipping: $", totalCost

ELSE IF (packageWeight <= PACKAGE MIN)

OUTPUT "Invalid package weight"

ELSE IF (packageWeight > PACKAGE MAX)

OUTPUT "PACKAGE EXCEEDS WEIGHT LIMIT"

END

Implementation and Githutb

The program is organized by first taking input, then a nested if else statement. The statement accounts for invalid cases before checking the package weight and shipping distance within the expected range.

I start off by taking the user input and storing in a variable, I check its working as expected by outputting them. Then I begin checking the user input for package weight requirements. After that I add shipping distance requirements, including invalid statements for both. Then I add some output statements that include the multiplier to do testing. Once the testing was completed, I added in constants and cleaned up the output with clear labels.

Testing

Weight	Distance	Test Purpose	Expected Result	Actual Result	Pass or Revision
-1	10	Package weight < 0	Invalid package weight	Invalid package weight	Pass
21	10	Package weight > 20	Package exceeds limit	Package exceeds limit	Pass
2	6	Shipping distance < 10	Invalid shipping distance	Invalid shipping distance	Pass
2	3001	Shipping distance > 3000	Invalid shipping distance	Invalid shipping distance	Pass
5	600	Non-multiple of 500 for distance	$4.40	$4.40	Pass
6.5	1700	Double value for weight, 3rd weight range	$14.80	$14.80	Pass
16	1000	Multiple of 500, 4th weight range	$9.60	$9.60	Pass

Testing Summary

The test results demonstrate the selection logic, calculations, and output are correct by first checking for valid package weights and then checking for a valid shipping distance. Then a non-multiple for 500 is tested to check the multiplier rounds up as expected. The next test checks a double as user input for weight and the 3rd weight range. The last test checks for a standard multiple of 500, 3 x multiplier, and 4th weight range.

Reflection:

1. The boundary difficult to implement was the package weights the second time because of the nested if statements,
2. Rounding upward was necessary because any amount of miles over increments of 500 needed to have the number for the rate that rounds up as well.
3. The 6th test demonstrated the program handled exact and non-exact multiples of 500 correctly by having a use input of 1700 for shipping distance.
4. I added the ceil function before I added the cmath library at the top of the code.
5. Formatting the result for currency greatly improves the readability and accurave of the program.